

MACAD 2025/26 · DIGITAL TOOLS FOR GRAPH MACHINE LEARNING

GraphML_Go2

Graph Learning Pipeline for Architectural Floor Plans

FINAL ASSIGNMENT

IAAC · MaCAD 2025/26

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Contents

This week we explored the Swiss Dwellings dataset and the research papers behind it, and defined the next steps for our graph ML pipeline.

01 Dataset

Modified Swiss Dwellings — 5,372 residential floor plans as graphs · Plan 10000 selected

02 Research

MSD benchmark paper (ECCV 2024) · GNN node classification paper (eCAADe 2024)

03 Graph Structure

graph_in vs graph_out — zone labels, connectivity, and room type prediction

04 Next Steps

Floor plan recreation → graph construction → analysis → node classification

01 DATASET

Modified Swiss Dwellings

- 5,372 floor plans
- Multi-apartment clusters
- Corridor buildings
- Single units
- Graph modality only



3 sample plans — wide variety of residential typologies, scales and orientations

5,372

Floor Plans

4

Zone Types

9

Room Types

.pickle

Graph Format

01 DATASET — REFERENCE PLAN

Plan 10000 — Dataset Reference



41 Rooms
across 2 apartments

2 Apartments
sharing a common entrance

→ **Our Plan**
we will design our own residential floor plan in Rhino

Cluster typology Residential

Plan 10000 — used to understand the dataset structure and typology

02 RESEARCH

Literature

PAPER 1

MSD: A Benchmark Dataset for Floor Plan Generation of Building Complexes

ECCV 2024 · TU Delft

- 5,372 Swiss residential floor plans
- 3 modalities: image, geometry, graph
- Richer than RPLAN & LIFULL — multi-apartment, compass orientation, inter-unit links
- Graph encodes spatial zones, not room types

This dataset is our primary data source.

PAPER 2

GNN for Node Classification and Attribute Allocation in Architectural BIM

eCAADe 2024 · Cardiff University

- Uses the same MSD dataset
- Converts floor plans to graphs via TopologicPy
- Benchmarks GCN, GAT, GraphSAGE architectures
- GraphSAGE-Pool, 4 layers — ~95% accuracy

The pretrained model from this paper is what we will run on our floor plan.

03

GRAPH STRUCTURE

Data Structure

graph_in — model input

Attribute	Type	Values
zoning_type	int	0 · 1 · 2 · 3
connectivity	str	'door' · 'entrance'

graph_out — ground truth

Attribute	Type	Description
room_type	int	0 – 8
geometry	Polygon	2D room outline
centroid	Point	centre of room

Room types (0 – 8)

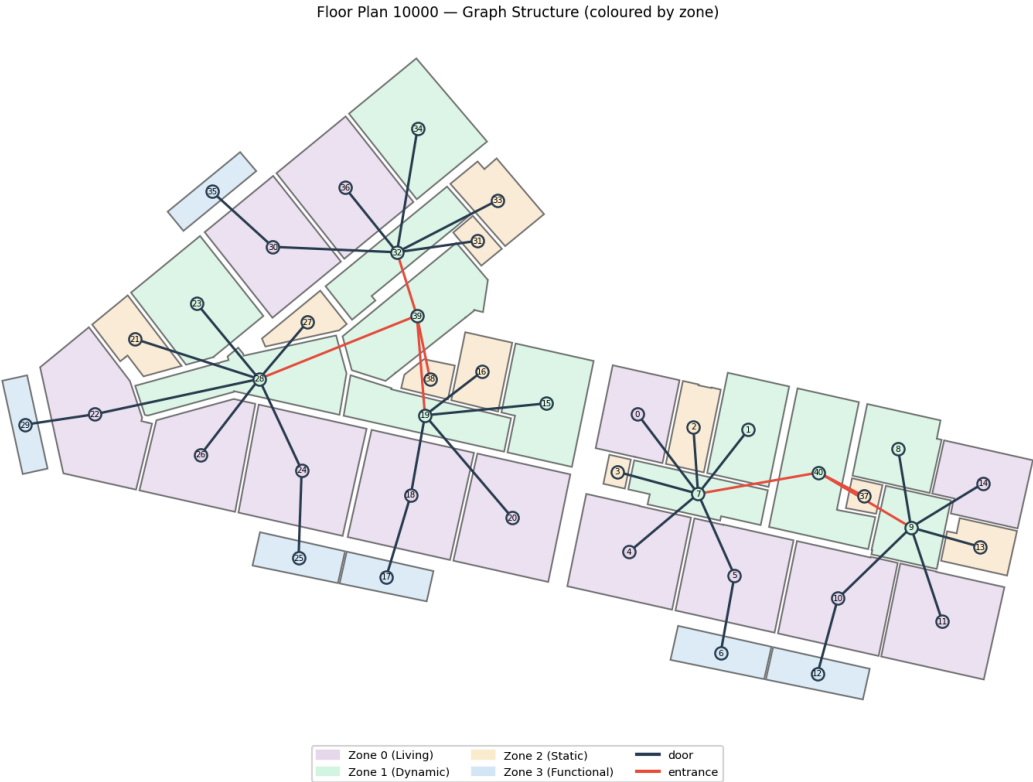
ID	Room	ID	Room
0	Balcony	5	Kitchen
1	Bathroom	6	Living Room
2	Bedroom	7	Storeroom
3	Corridor	8	Other
4	Dining		

Zone types (0 – 3)

ID	Zone	Typical rooms
0	Living	Living room, dining
1	Dynamic	Corridor, entrance
2	Static	Bedroom, bathroom
3	Functional	Kitchen, storeroom

03 GRAPH STRUCTURE

From Plan to Graph — Plan 10000



Plan 10000 — room types (left) and spatial connectivity graph (right)

Zone label per node

graph_in input



Graph topology

door / entrance edges



Room type prediction

GraphSAGE-Pool output

04 NEXT STEPS

Pipeline

Floor Plan Design

- 01** Design our own residential floor plan in Rhino. Room boundaries as closed polylines with shared edges between adjacent rooms — required for TopologicPy to detect adjacency.

Graph Construction — TopologicPy

- 02** Convert the Rhino geometry into a graph. Assign zoning_type to nodes and door/entrance labels to edges to match the MSD input format.

Graph Analysis — TopologicPy

- 03** Compute degree, betweenness, and closeness centrality using TopologicPy. Analyse spatial connectivity patterns across the floor plan.

Node Classification — GraphSAGE-Pool

- 04** Run the pretrained model on our graph. Predict room_type from zone labels and connectivity. Compare against ground truth from MSD.